Creating Trauma-Informed Research Technologies Through Dating App Sexual Violence Research

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Abstract

The Trauma-Informed Approach (TIA) is a framework from the health and human services sphere for interacting with trauma survivors that is widely used in the social computing community to mitigate harm to technology users. The framework provides four assumptions and six principles that show promise for mitigating retraumatization, defined as reflections on traumatic events that harm an individual's wellbeing. My PhD explores the application of the trauma-informed approach to social computing research technologies in the context of online-to-offline sexual violence, such as when a person meets someone on a dating app and experiences sexual violence. My research and subsequent publications provide guidance for other researchers and designers who collect data that may be related to experiences of trauma.

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1 Introduction

Trauma, or negative impacts on an individual's well-being as a result of experiences such as disaster, violence, or abuse [8], is a common focus in social computing research as the motivation for (e.g., [1, 12, 14]) and topic of study (e.g., [18, 24]). Social computing thus far examines incidents that can lead to trauma such as online dating sexual violence and/or consent [2, 7, 23], bullying [5, 10], and child abuse [3, 6], among other harms.

Trauma mitigation in social computing is most often pursued through application of the Trauma-Informed Approach (TIA)–a theoretical framework created by the Substance Abuse and Mental Health Services Administration (SAMHSA) [8] in the United States. Outside of social computing, TIA intends to guide organizations and trauma care workers on how to provide services for individuals who have experienced trauma. TIA recognizes and mitigates retraumatization: when a person relives "stress reactions experienced as a result of a traumatic event when faced with a new, similar incident" [8]. Other social computing researchers have sought to translate TIA to computing through the proposal of Trauma-Informed Computing principles [4] and trauma-informed design heuristics [9], for instance, and by applying TIA to specific contexts of trauma [1].

Notably, trauma-informed technologies vary in how they qualify themselves as adhering to the trauma-informed approach. Saxena and Gudha, for instance, consider their algorithm for child welfare to be trauma-informed due to their end goal, which is to better understand parents and children, ending with empowerment for users [15]. This differs from a common tactic in social computing literature that uses trauma experts as proxies for actual users [1, 11, 19]. In another vein, Scott et al. [16] reflect on how the principles of TIA may be best applied to social media. These differential methods for applying the trauma-informed approach broaches questions around what should be considered trauma-informed, and what should *not*.

My dissertation explores application of the Trauma-Informed Approach to social computing research technologies in an effort to mitigate retraumatization to participants within social computing research. I bridge the gap between trauma-informed experts, users of the technologies, and social computing researchers, among other disciplines. I conduct this research in the context of sexual violence related to dating apps, when a person meets an individual in-person whom they originally met on a dating app, and then experiences sexual violence. This is a great context for trauma-informed research due to multiple established ethical concerns within sexual violence research such as risk of incidentally labeling a participant a victim or perpetrator of sexual violence who does not consider themselves as such [13], and the involvement of more than one individual across all dating app interactions.

2 Background

2.1 The Trauma-Informed Approach

The Trauma-Informed Approach (TIA) originated in the health and human services sphere as guidance for healthcare experts who interact with those who have recent trauma experiences. The framework provides four assumptions and six principles that all providers should adhere to when administering care [8].

The four assumptions of TIA assert that trauma-informed care should (1) *Realize* the impact of trauma, (2) *Recognize* signs and symptoms of trauma, (3) *Respond* to trauma with principles of TIA, and (4) *Resist* retraumatization.

The six principles of TIA prompt those administering trauma-informed care to consider (1) *Safety* of those they work with, (2) *Trustworthiness and transparency* between those who provide and receive care, (3) *Peer support* by others who have experiences of



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trauma, (4) *Collaboration and mutuality* by leveling power differences between those administering and receiving care, (5) *Empowerment, voice, and choice* by encouraging those who receive care to make their own decisions, and finally (6) *Cultural, historical, and gender issues* by avoiding stereotypes and understanding and applying knowledge around intersectionality.

3 Research Goals

My dissertation and publications will provide social computing researchers with guidance on how they may prevent retraumatization in studies that collect data that may be related to experiences of trauma. There are four research questions driving my dissertation:

(1) **RQ1**: How do we qualify social computing technologies as trauma-informed?
(2) **RQ2**: "Whose" trauma: which stakeholders should be incorporated in trauma-informed design? - it's not just the immediate user
(3) **RQ3**: How do we involve both trauma experts and recipients of the trauma-informed technology in evaluating adherence to TIA?
(4) **RQ4**: How can trauma-informed social computing research technologies be informed by trauma-informed trainings at trauma care clinics?

4 Methodology

Research Through Design (RTD): To explore RQ1 and RQ3, I conducted a 15-month RTD process to iteratively design and develop an app for donation of data (a method that allow donors to provide screenshots of their interactions) about online-to-offline sexual experiences related to dating apps. The RTD process included experts from the fields of Forensic Nursing, Computing, Psychology, Software Development, and Social Work. Trauma-informed methods and technology developed throughout this process were used in the subsequent observational interviews.

Observational interviews: To answer RQ2, I designed and analyzed a study where donors, who were frequent dating app users, used the trauma-informed app designed using RTD to donate data related to online dating experiences. Donors, either naturally or after prompting, reflected on their willingness and comfort to provide screenshots of interactions that included more than one individual's text messages and/or images.

Ethnographic exploration of trauma clinics: I am in the midst of an ethnographic study to answer RQ4. I am currently in the middle of participating in a training process for the role of an advocate at a trauma-informed clinic who specifically assists survivors of sexual violence, domestic violence, and human trafficking. This will not be the only training I participate in; I plan to attend trainings at three to four other trauma-informed clinics as well.

5 Work Completed to Date

Here, I will describe the three papers I have published and one paper that has been accepted for publication. These comprise published results of the first three RQs of my dissertation.

First, my team's 2024 CHI full-paper [22] and preceding poster paper in UIST 2023 [21] (both of which I am co-first author) demonstrate published results of our 15-month Research Through Design process to design and develop a trauma-informed data donation [17] app for donation of online-to-offline sexual experiences. Here, we explore how design teams apply TIA and qualify their technology as adhering to trauma-informed principles. Ultimately, we propose three related concepts to fuel trauma-informed design processes: design goals to adapt TIA definitions to application, design activities that directly map to TIA principles, and finally design choices that result from these goals and activities.

The next paper within my dissertation is published in CSCW 2025 [20]. This paper reports on an observational interview study utilizing the trauma-informed method and designed technology resulting from RTD. In this study, data donors reflected on their willingness and comfort to provide screenshots and short-answer responses that involve interactions with other individuals while they donated data using the trauma-informed platform. This study reminds us that trauma is often a social experience; when we study harms that are social in nature, there is more than one individual to consider when digital trace data (e.g., screenshots of messaging interactions) is collected. Donors in this study in particular were especially concerned about maintaining relationships with partners who they got along with, and worried about their own safety when providing data that involved a perpetrator of sexual violence.

I also have one paper accepted that has not yet been published. It is titled "Trauma-Informed Data Donation: Integrating Expert and Donor Perspectives on Designing Against Retraumatization During Collection of Sexual Violence Data." This paper will be published in DIS 2026 in June. In this work, I explore how social computing can involve two differing and yet essential perspectives on application of the trauma informed approach: donors who may have experiences of trauma, and trauma experts. The paper reports on our 2-year RTD process; it elucidates the specific impacts made by both donors and users, and helps us see the value and challenges in including both perspectives. It also provides a framework for trauma-informed data donation that can inform other scholars who plan to conduct data donation on topics that may be related to experiences of trauma.

6 Purpose of Doctoral Consortium

At this point, I have completed the first three research questions outlined in Section 3. I plan to propose my dissertation in the late summer of 2026 and hope to defend my dissertation in early 2027.

This doctoral consortium comes at a vital time as I am formulating the ethnographic study that I will conduct to answer RQ4. I am new to ethnographic methods, and I want to ensure that I have a valuable plan for conducting the research, and for what a valuable contribution from this study would look like within the social computing community.

I have a MS degree in experimental psychology, and since transitioning to HCI, I am still finding my footing in terms of understanding when a study is valuable and why. Discussing with scholars who study similar things would help me grow as a researcher by helping me learn how to ensure (1) my dissertation is of the highest quality it can be in a social computing space, and (2) I continue contributing to social computing in a valuable way.

7 Long-Term Goals

My dissertation will be composed of the aforementioned research projects. At this point, three of four of my research questions have been answered (RQ1, RQ2, RQ3). Before I finish my PhD, I will finish conducting my ethnographic study that encapsulates RQ4, and I will publish the findings for the social computing community. I aim to defend my dissertation at the end of the coming academic year.

Long-term, I aim to remain in academia continuing research on similar topics to these. I plan to continue research on trauma-informed methods and effective mitigation of retraumatization in research.

Importantly, these trauma-informed methods led to a valuable dataset of online-to-offline sexual data that can be used in my future work. Additionally, my future research may include continued collection of data on technology-facilitated sexual violence from groups such as LGBTQ people and those who are not in the US. I am in a unique position to collect data like this and do so without harming participants due to my familiarity with application of TFSV as gained throughout my dissertation studies, thus my PhD studies will set me up to thrive in an academic position and to continue to contribute to the CSCW community.

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